

Matthew Leung

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EDUCATION

University of California, San Diego

La Jolla, CA

Master of Science in Computer Engineering

September 2026 – Present

Bachelor of Science in Computer Engineering

September 2022 – June 2026

- GPA: 3.98

EXPERIENCE

Electrical Engineering/Cyber Security Intern

June 2026 – Present

TrustThink

- Performed penetration testing on unmanned aerial systems across firmware, wireless interfaces, and physical ports
- Uncovered and documented 5+ high/medium-severity vulnerabilities, including unsigned protocols, unencrypted filesystems, and exposed debug ports
- Coordinated with 5–10 engineers daily on security reporting and sprint planning while communicating with clients on vulnerability remediation
- Exploited MAVLink vulnerabilities through message-spoofing scripts and physical port manipulation

Computer Science Tutor (CSE 30)

January 2026 – March 2026

University of California, San Diego

- Led weekly discussion sections for 15+ students covering computer architecture concepts, including caching, stack frames, and ARM calling conventions
- Hosted weekly office hours for 3–5 students, providing hands-on debugging support with GDB and feedback on exam performance
- Designed instructional materials and collaborated with tutors to deliver course-aligned support

PROJECTS

Junkyard Computing – OTA Updates | *Linux, Bash, dracut, btrfs*

March 2026 – June 2026

- Contributed to a Google-sponsored research project developing scalable compute clusters from repurposed Google Pixel devices, focusing on OTA update infrastructure
- Engineered an A/B rootfs update system using btrfs subvolumes to safely deploy and roll back OS updates
- Automated rootfs versioning and boot-time update management in a custom dracut module
- Designed an update staging system that moves rootfs storage from the 8 GB super partition to userdata, expanding available storage by up to 25x

RealTime Display | *ESP-32, C++, FreeRTOS, Python, Flask, Onshape*

September 2025 – December 2025

- Built an all-in-one LED display with a clock, weather monitor, Spotify visualizer, and live sports tracker
- Implemented FreeRTOS mutexes and task synchronization to manage concurrent network fetching, rendering, input handling, and display updates
- Implemented a modular client-server architecture for creating and adding custom widgets
- Designed and fabricated a 3D-printed enclosure with physical user controls

Lava LiDAR Game | *Arduino, Python, C++*

April 2025 – June 2025

- Integrated a LiDAR sensor with an Arduino Uno via I2C to enable real-time player tracking in an interactive game
- Developed a serial communication pipeline between the Arduino and Python game engine for responsive motion control
- Awarded 1st place at UCSD's IEEE Quarterly Projects Showcase (Spring 2025)

TECHNICAL SKILLS

Languages: Java, Python, C/C++/C#, SystemVerilog, GLSL, SQL (Postgres), TypeScript, JavaScript

Frameworks/Runtime Environments: React, Node.js, Bootstrap, Flask, JUnit

Developer Tools: Git, MongoDB, Google Cloud Platform, Hugging Face, Wireshark, Ghidra

Communication Protocols: UART, CAN, I2C, MAVLink

Hardware Skills: Digital systems design, LTspice, ModelSim, Cadence Virtuoso, FreeRTOS